

PathNER AI 2.0

Smart Logistics & Dynamic Accessibility Platform for Northeast India

Comprehensive System Architecture, Operational Workflow, and Presentation Reference Manual

PathNER is an end-to-end AI-powered geospatial intelligence and supply-chain resilience platform engineered specifically for the fragile mountain topography of Northeast India (Assam, Meghalaya, Dima Hasao, Barak Valley). It safeguards critical lifeline arteries by combining Machine Learning hazard forecasting, real-time graph reachability, multi-criteria safe routing, generative multilingual advisories, and offline PWA field reporting.

Event / Project: Smart India Hackathon (SIH 2026)

Team: UpWorld Devs

Target Geography: Northeast Region (Guwahati, Shillong, Haflong, Silchar, East Jaintia Hills)

Datasets Processed: 968 Settlement Nodes & 824 Authentic Road Segments (OpenStreetMap + Topographic Elevation)

Core AI Engine: Pre-Trained Risk Model (Gradient Boosting / Random Forest) + Mistral AI Tactical LLM

Offline Engine: ServiceWorker + LocalStorage Queue (PWA)

1. Executive Summary & Problem Context

The Challenge in Northeast India:

The Northeast Region (NER) of India is geologically active and experiences some of the highest precipitation in the world (e.g., Cherrapunjee and Mawsynram). During monsoons, strategic national highways—especially **NH-6 (connecting Assam, Meghalaya, and Barak Valley)** and **NH-27 / NH-627 (through Dima Hasao)**—suffer frequent, catastrophic disruptions from mudslides, boulder falls, and flash floods. When an arterial road collapses:

- **Severed Supply Chains:** Remote communities run out of food, medicine, and fuel within 3 to 5 days.
- **Blind Dispatch Routing:** Standard GPS navigation apps (like Google Maps) blindly route heavy relief trucks into impassable active landslide zones.
- **Zero-Signal Field Blindspots:** Ground engineers from BRO and PWD lack cellular connectivity in steep hill valleys to transmit incident reports.

The PathNER Solution:

PathNER solves these three vulnerabilities through a unified, 4-pillar digital platform:

1. **ML Multi-Variable Hazard Prediction:** Pre-trained Gradient Boosting models compute real-time risk scores (0–100%) for every road corridor using terrain slope gradient, soil saturation, and precipitation forecasts.
2. **Dynamic Graph Reachability (Isolation Index):** The platform runs continuous Breadth-First-Search (BFS) reachability scans across all 968 nodes to identify communities cut off from supply hubs before emergency supplies deplete.
3. **Multi-Criteria AI Route Dispatcher:** Computes dual route options: the dangerous standard path vs. the AI-Recommended Safe Bypass path that avoids high-risk bottlenecks.
4. **Offline-First Field Reporter (PWA):** Enables ground patrols to submit geo-tagged incident reports without cellular network, queueing them in local storage and auto-syncing upon reconnection.

2. Technical Architecture & Data Pipeline

Component Layer	Technology Used	Core Responsibility
Backend Server	Python 3.12, Flask 3.0	REST API endpoints, routing calculations, session cache, sub-millisecond graph evaluations.
ML Risk Engine	scikit-learn (pkI model)	Pre-trained multi-factor hazard estimation combining slope (deg), rainfall (mm), soil saturation, and active field reports.
Graph Engine	Custom GraphEngine (BFS / Dijkstra / A*)	968-node & 842-edge road network, dynamic impedance weighting, reachability graph evaluation.
Tactical LLM	Mistral AI API (with 10m TTL cache)	Generates military-grade disaster advisories in English, Assamese, Bengali, and Hindi with zero credit waste.
Database	SQLite 3 (WAL mode)	Field reports, multilingual emergency alerts, user notification subscriptions with idempotent UUIDs.

Frontend & GIS	Leaflet.js, Vanilla CSS / JS (Dual-Theme)	High-contrast dual-stroke road rendering, 4 basemaps, interactive layer filter HUD, 100% mobile responsive.
Offline Edge	PWA / LocalStorage / ServiceWorker	Stores field reports locally in zero-signal mountain zones and auto-syncs when online.

3. End-to-End Module & Workflow Walkthrough

Module 1: Control Room Command Center (/)

What it does: The central executive operations dashboard giving commanders an instant bird's-eye view of regional logistics health.

- **Live Top KPI Cards:** Real-time counters showing *Stranded Population* (fully cut off), *Severed Hamlets* (0% reachability), *At-Risk Population* (impending isolation), and *Blocked Corridors* ($\geq 70\%$ risk).
- **72-Hour Timelapse Simulation Player (■):** Click the Play button to watch an animated 4-step forecast progression (Now → +24h → +48h → +72h) showing how corridors degrade as monsoon rains accumulate.
- **Interactive Layer Filter HUD:** Toggle buttons (Roads, Hubs, Chokepoints, Fleet) allowing dispatchers to declutter map information.
- **Live Fleet Simulator:** Tracks 4 autonomous relief convoys navigating live delivery routes.

Module 2: AI Multi-Criteria Route Dispatcher (/routes)

What it does: Calculates the optimal delivery path between central supply depots and remote delivery points.

- **Side-by-Side Path Comparison:** Simultaneously evaluates the *Standard Shortest Path* (fastest under normal conditions, but traverses active landslides) versus the *AI-Recommended Safe Bypass* (minimizes slope hazard and bypasses breaches).
- **AI Decision Verdict Banner:** Plain-language summary explaining exact trade-offs (e.g., "*AI Reroute Recommended: Safe bypass adds 42km (+0.9h) but avoids 1 active landslide bottleneck, reducing mission risk from 74% to 18%.*")
- **Turn-by-Turn Waypoint Stepper:** Displays intermediate waypoints, elevation gradient, and clearance alerts for convoy drivers.

Module 3: Field Incident Reporter & Offline PWA (/field-reporter)

What it does: Mobile-first digital reporting interface for BRO patrol officers and PWD engineers on the ground.

- **Offline Queue Architecture:** In zero-connectivity valleys, clicking 'Transmit Report' saves the incident into browser LocalStorage. As soon as the patrol vehicle enters 2G/4G coverage, the PWA auto-syncs the queue to Central Command.
- **Live Judge Simulation Bar:** Built specifically for hackathon evaluators:
 1. Click 'Load Demo Incident' → auto-fills a realistic BRO rockfall report with exact GPS coordinates.
 2. Click 'Simulate Offline Mode' → toggles zero-signal state; submit report to queue locally.
 3. Click 'Restore Online & Sync' → triggers background sync, updating the central map in real-time.

Module 4: Tactical Emergency Response Mode (/disaster-mode)

What it does: High-urgency command mode activated during active flood/landslide disasters.

- **AI Tactical Directive Console:** Generates real-time logistical directives using Mistral AI, translated on-the-fly into 4 languages (English, Assamese, Bengali, Hindi). Responses are cached both on backend (10min) and frontend to save 100% of repeated API credits.
- **Severed Settlements Priority Queue:** Lists cut-off hamlets ranked by urgency, showing remaining food/medicine buffer days (countdown bar) and population.
- **1-Click Relief Pre-fill:** Click 'Dispatch Relief →' on any severed settlement to instantly open the Route Dispatcher with that destination pre-selected.

Module 5: Drishti AI — Conversational Logistics Copilot (Floating Drawer)

What it does: An interactive floating AI assistant (named **Drishti AI** — *Predictive Vision & Foresight*) accessible across every page.

- **Live Platform Awareness:** Automatically injects current real-time telemetry into every conversation (active road blocks, landslide hazards on NH-6, severed settlements count, and convoy status).
- **Dual-Engine Architecture:** Powered by Mistral AI LLM with an intelligent local domain NLP fallback engine, delivering sub-second replies with zero API credit waste.
- **Multilingual Natural Language:** Understands and answers disaster inquiries in English, Hindi, Assamese, and Bengali.
- **1-Click Quick Chips:** Allows emergency commanders to instantly query corridor status ('■■■ NH-6 Status', '■■■ Route to Silchar', '■ Isolated Hamlets', '■ Offline Reporting').

4. Presentation Pitch & Judge Demonstration Guide

Use this structured 3-minute pitch flow when presenting PathNER to judges or team members:

Step & Time	What to Say (Script in Simple Words)	What to Show on Screen
1. The Hook (0:00 - 0:40)	"Every monsoon in Northeast India, landslides on NH-6 cut off the entire Barak Valley and Meghalaya hills. When roads collapse, thousands of people run out of food and medicine in 3 days. Google Maps cannot tell you if a mountain road is collapsing. That's why we built PathNER AI."	Open Control Room (/) . Point out the Live KPI cards: 4 Severed Hamlets, 13,453 Stranded Population, and the clean map with high-contrast road corridors.
2. AI Routing (0:40 - 1:20)	"Watch what happens when a dispatcher routes supplies from Guwahati to Silchar. Standard apps suggest the shortest path through Jaintia Hills, which is currently blocked by a 74% risk landslide. PathNER's AI algorithm automatically calculates a safe bypass via Haflong, adding a few kilometers but ensuring 100% mission safety."	Navigate to Route Dispatch (/routes) . Click the preset ' <i>Guwahati → Silchar</i> '. Show the orange dashed blocked route vs. the solid emerald safe route and the Decision Banner.
3. Offline PWA (1:20 - 2:00)	"In the steep hills, road engineers have zero cell signal. Our Field Reporter works 100% offline. An officer submits a report in a dead zone, it queues in the browser, and as soon as signal returns, it automatically syncs with the Control Room, updating the entire AI reachability model."	Open Field Report (/field-reporter) . Click ' <i>Load Demo Incident</i> ' → click ' <i>Simulate Offline Mode</i> ' → submit → show offline queue card → click ' <i>Restore Online & Sync</i> '.
4. Emergency Mode (2:00 - 2:30)	"In Emergency Mode, our Mistral AI tactical engine delivers multilingual briefings in Assamese, Bengali, Hindi, and English, prioritizing communities whose supply buffer is about to hit zero."	Open Emergency (/disaster-mode) . Toggle the language buttons (EN / Assamese / Bengali / Hindi) to show instantaneous translated directives.
5. Drishti AI Demo (2:30 - 3:00)	"Lastly, commanders can chat directly with our Drishti AI Copilot in natural language. Click a quick prompt chip or ask 'Is NH-6 open?', and it immediately gives live, contextual answers."	Click the Drishti AI button in the bottom right corner → click ■ ■ NH-6 Status chip → show instant tactical intelligence reply.

Key Differentiators & Competitive Advantage:

- 1. Real Localized Data:** Uses 968 real GPS nodes & 824 authentic OpenStreetMap corridors from Northeast India.
- 2. Sub-Millisecond Speed:** In-memory graph caching evaluates entire regional networks in under 1ms.
- 3. Dual-Theme High Contrast:** Perfectly readable in both sunlight/light mode and night tactical dark mode.
- 4. Offline-First Resilience:** Zero cell signal requirement for field responders.
- 5. Drishti AI Copilot:** Real-time conversational AI copilot for emergency commanders.